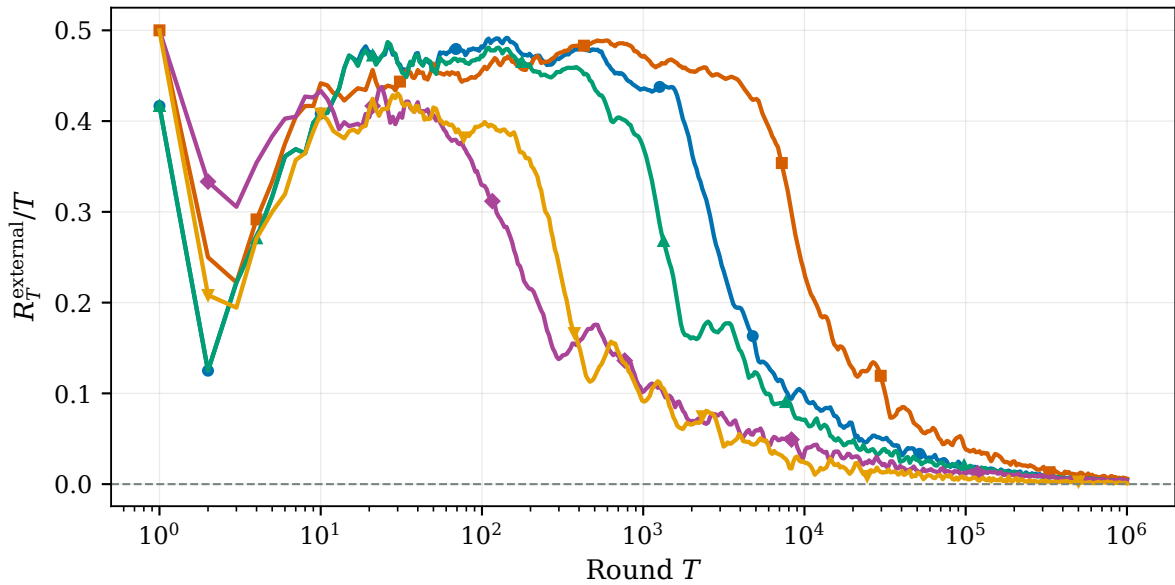


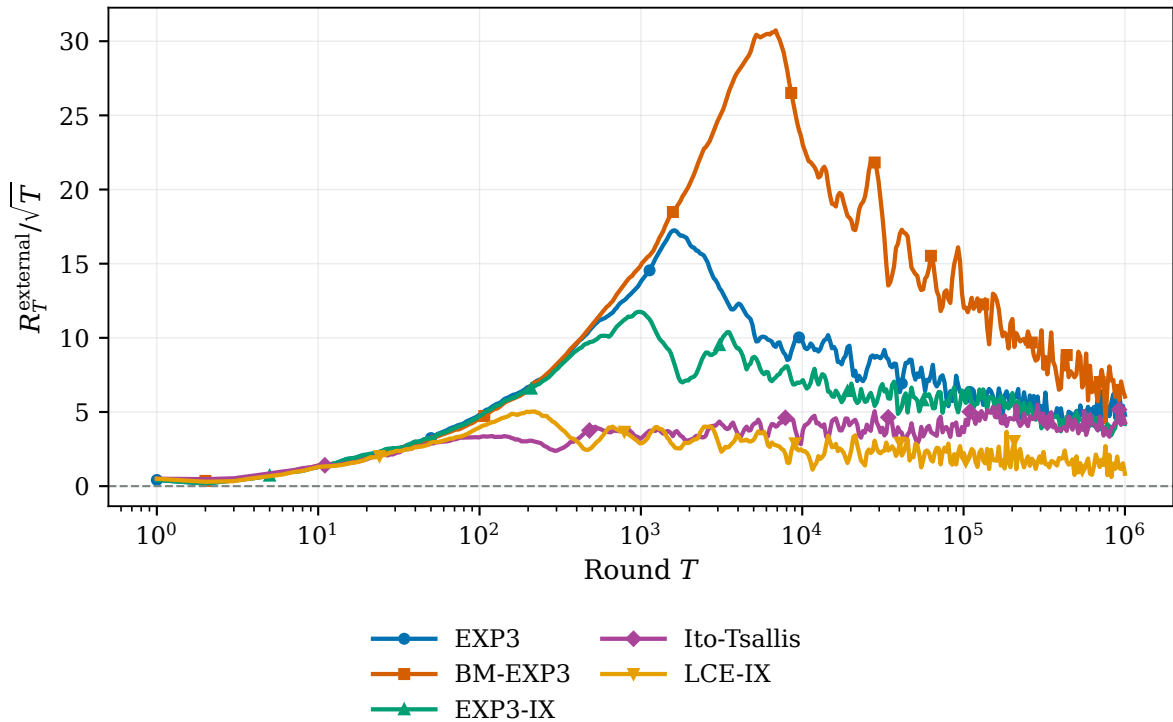
Experiment Information

Figure: Algorithm comparison
Environment: Historical-frequency
Actions: 10
Feedback: Bandit feedback
Algorithms: EXP3, BM-EXP3, EXP3-IX, Ito-Tsallis, LCE-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3: mean = 0.004803, SD = 0.00170114, SE = 0.000491078
BM-EXP3: mean = 0.00603992, SD = 0.00179243, SE = 0.000517429
EXP3-IX: mean = 0.00419725, SD = 0.0025064, SE = 0.000723535
Ito-Tsallis: mean = 0.0042845, SD = 0.00121424, SE = 0.000350521
LCE-IX: mean = 0.000834583, SD = 0.00119436, SE = 0.000344781



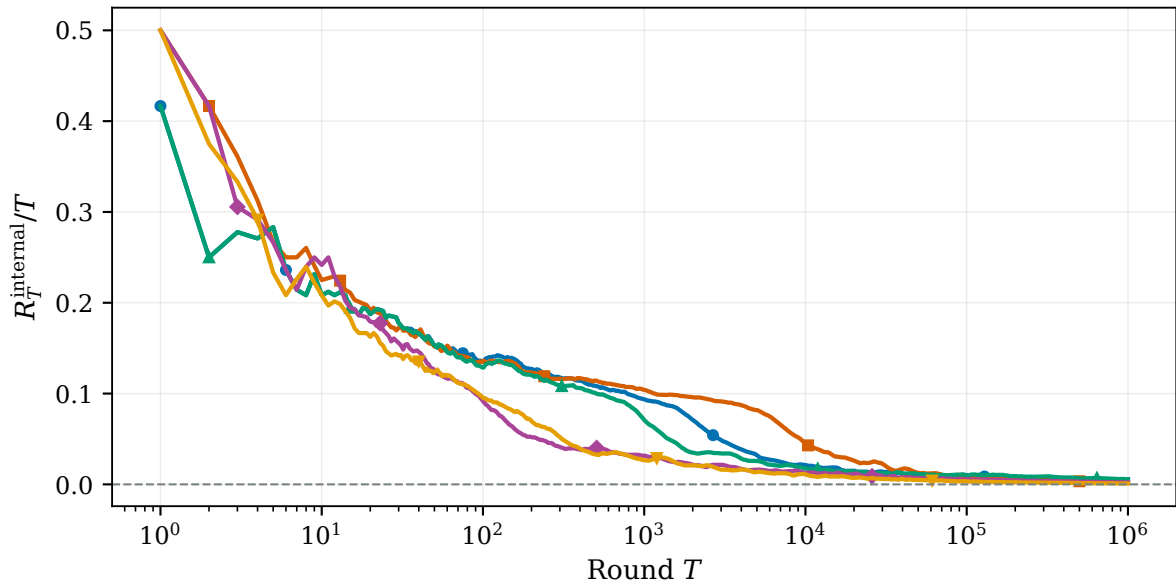
Experiment Information

Figure: Algorithm comparison
Environment: Historical-frequency
Actions: 10
Feedback: Bandit feedback
Algorithms: EXP3, BM-EXP3, EXP3-IX, Ito-Tsallis, LCE-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3: mean = 4.803, SD = 1.70114, SE = 0.491078
BM-EXP3: mean = 6.03992, SD = 1.79243, SE = 0.517429
EXP3-IX: mean = 4.19725, SD = 2.5064, SE = 0.723535
Ito-Tsallis: mean = 4.2845, SD = 1.21424, SE = 0.350521
LCE-IX: mean = 0.834583, SD = 1.19436, SE = 0.344781



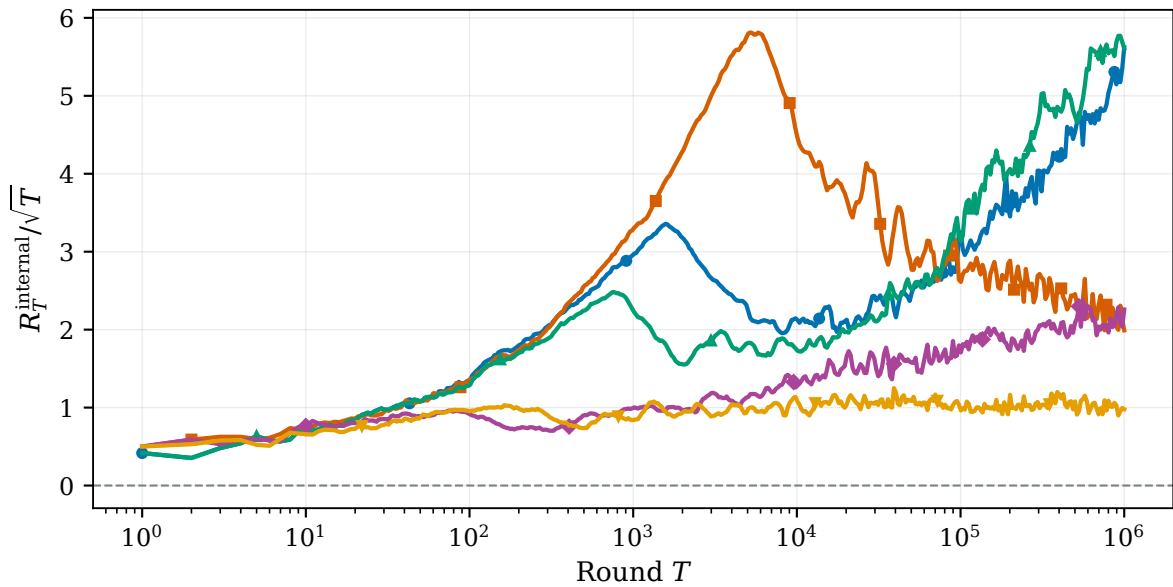
Experiment Information

Figure: Algorithm comparison
Environment: Historical-frequency
Actions: 10
Feedback: Bandit feedback
Algorithms: EXP3, BM-EXP3, EXP3-IX, Ito-Tsallis, LCE-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3: mean = 0.00562192, SD = 0.00148798, SE = 0.000429542
BM-EXP3: mean = 0.00199142, SD = 0.000436228, SE = 0.000125928
EXP3-IX: mean = 0.00560775, SD = 0.00108382, SE = 0.000312871
Ito-Tsallis: mean = 0.00225817, SD = 0.000532163, SE = 0.000153622
LCE-IX: mean = 0.000978167, SD = 0.000230327, SE = 6.64896e-05



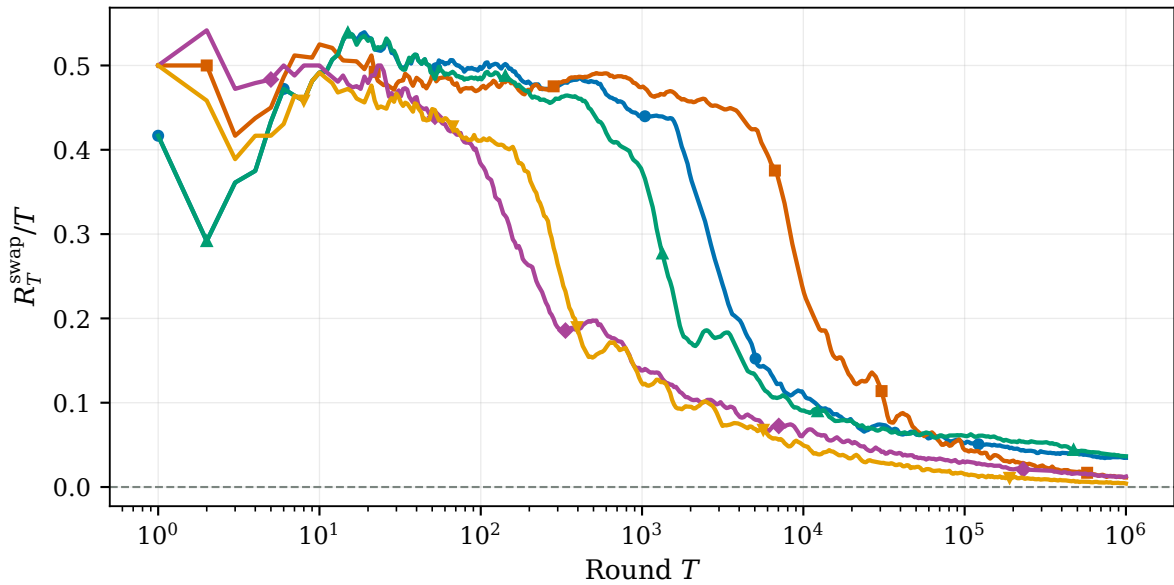
Experiment Information

Figure: Algorithm comparison
Environment: Historical-frequency
Actions: 10
Feedback: Bandit feedback
Algorithms: EXP3, BM-EXP3, EXP3-IX, Ito-Tsallis, LCE-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3: mean = 5.62192, SD = 1.48798, SE = 0.429542
BM-EXP3: mean = 1.99142, SD = 0.436228, SE = 0.125928
EXP3-IX: mean = 5.60775, SD = 1.08382, SE = 0.312871
Ito-Tsallis: mean = 2.25817, SD = 0.532163, SE = 0.153622
LCE-IX: mean = 0.978167, SD = 0.230327, SE = 0.0664896



Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Bandit feedback
Algorithms:	EXP3, BM-EXP3, EXP3-IX, Ito-Tsallis, LCE-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Swap regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3:	mean = 0.0346296, SD = 0.00799226, SE = 0.00230717
BM-EXP3:	mean = 0.0113095, SD = 0.00147638, SE = 0.000426195
EXP3-IX:	mean = 0.036513, SD = 0.00761423, SE = 0.00219804
Ito-Tsallis:	mean = 0.0124485, SD = 0.00251438, SE = 0.000725839
LCE-IX:	mean = 0.00415367, SD = 0.00111991, SE = 0.000323289



Experiment Information

Figure: Algorithm comparison
Environment: Historical-frequency
Actions: 10
Feedback: Bandit feedback
Algorithms: EXP3, BM-EXP3, EXP3-IX, Ito-Tsallis, LCE-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3: mean = 34.6296, SD = 7.99226, SE = 2.30717
BM-EXP3: mean = 11.3095, SD = 1.47638, SE = 0.426195
EXP3-IX: mean = 36.513, SD = 7.61423, SE = 2.19804
Ito-Tsallis: mean = 12.4485, SD = 2.51438, SE = 0.725839
LCE-IX: mean = 4.15367, SD = 1.11991, SE = 0.323289

